Smart Slip-Detection Glove for Prosthetic Hands

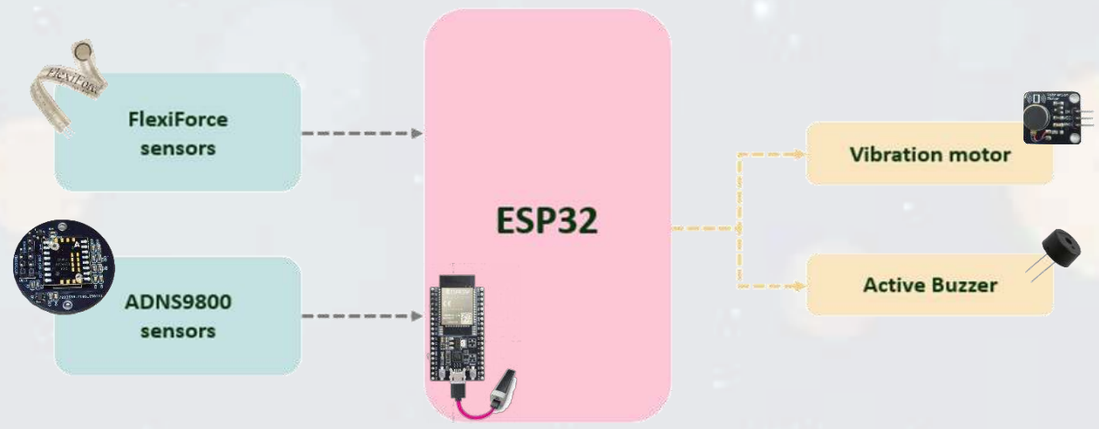
Ofir Goldstein and Gil Attar, Supervised by Koby Kohai
CRML, Faculty of Electrical and Computer Engineering, Technion

Motivation

Problem: Prosthetic hands cannot reliably sense when an object starts to slip, leading to dropped items and reduced user confidence. Existing solutions are expensive and complex, leaving a need for a simple, low-cost real-time slip-detection method.

Solution: Create a low-cost system using laser sensors and force sensors to detect slip reliably.

System Architecture

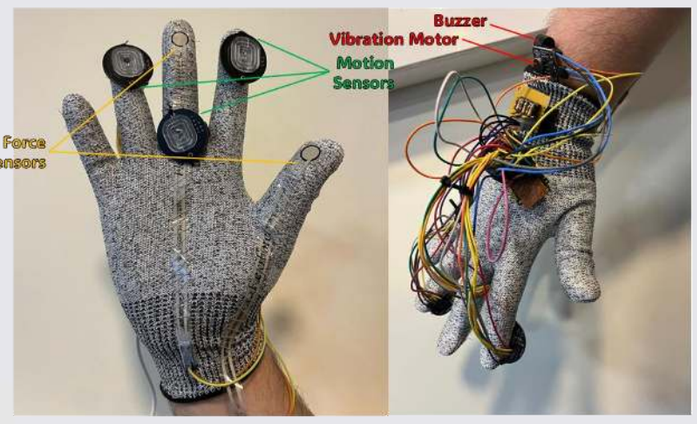


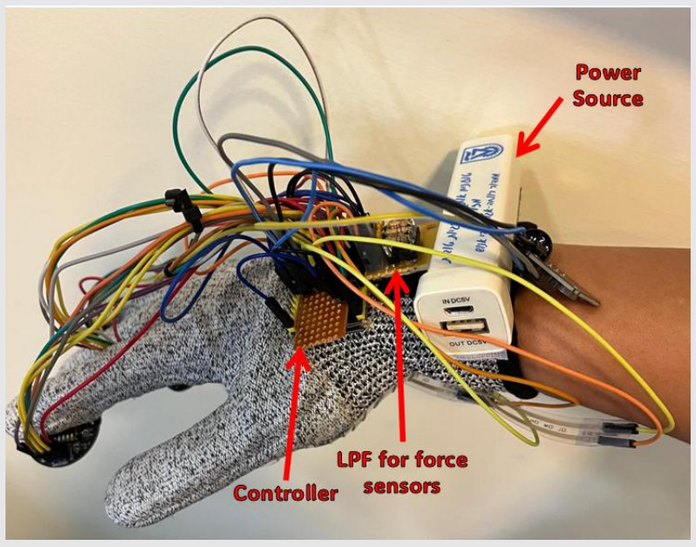
Laser and force inputs are processed together on the ESP32 to deliver fast, discreet slip-feedback to the user

Requirements

- Compact
- Reliable
- User Friendly Feedback
- Cost Effective
- Easy Integration
- Fast Response Time

Results

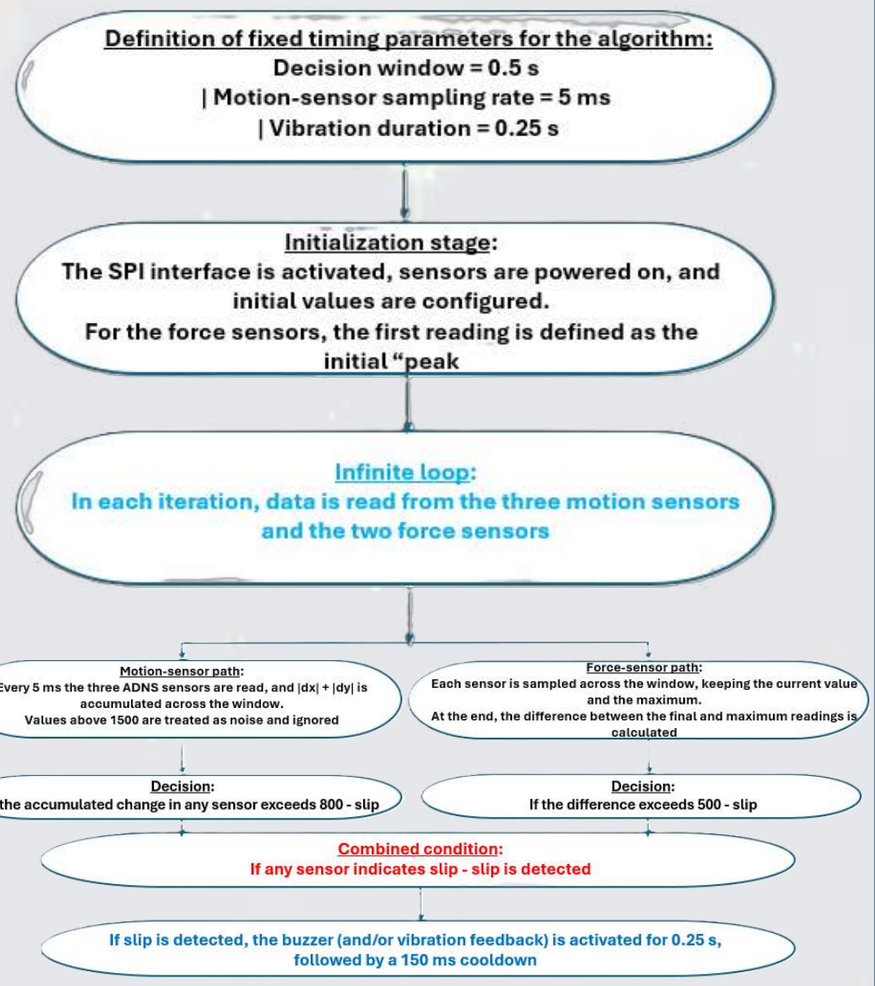




Results:

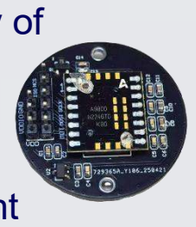
- Consistent slip detection across surfaces.
- Low false-positive rate.
- Fast and quiet vibration feedback.

System Algorithm



System Components

ADNS-9800 Laser Sensors
(~20–30$ each)



- Works on a wide variety of surfaces
- SPI interface
- Real-time slip-movement detection

FlexiForce Sensors
(~15–20$ each)



- Thin and flexible pressure sensors
- Measures grip-force changes
- Fast response, easy to integrate

ESP32 Microcontroller
(~10–15$)



- SPI + ADC interfaces
- Handles real-time processing and feedback

Others - Wires, connectors, Vibration motor (~10–15$)

Total System Cost: ~80–100$

Challenges

- Integration complexity when using raw sensor modules
  - ✓ Switched to PCB-mounted versions for reliability
  - ✓ Reduced wiring, noise, and mechanical instability
- Noise and instability in FlexiForce signals
  - ✓ Added low-pass filtering (RC)
- Motion asymmetry and large spikes in ADNS output
  - ✓ Implemented spike filtering (>1500)
  - ✓ Accumulated movement over a sliding window

Future Work

- Full integration into robotic prosthetic hand
- Adaptive thresholds / machine learning
- Mobile app for calibration
- Additional sensors (temperature, tactile arrays)